

LSTM Drug Demand Forecasting with Adjustment Strategies as a Preliminary Step Toward Optimizing Hospital Drug Inventory Management

Janejira LAOMALA ^a, Papon TANTIWANICHANON ^a, Natchanon JARUTEEKAMPRON ^a and Sayan KAENNAKHAM ^{b,1}

^a*Department of Interdisciplinary Science and Internationalization, Institute of Science, Suranaree University of Technology, Nakhon Ratchasima, 30000, Thailand*

^b*School of Mathematics and Geoinformatics, Institute of Science, Suranaree University of Technology, Nakhon Ratchasima, 30000, Thailand*

ORCID ID: Janejira Laomala <https://orcid.org/0009-0000-9350-2541>, Papon Tantiwanichanon <https://orcid.org/0009-0008-4336-8901>, Natchanon Jaruteekampron <https://orcid.org/0009-0004-1673-8701>, Sayan Kaennakham <https://orcid.org/0000-0001-9682-559X>

Abstract. Pharmaceutical inventory management plays a critical role in hospital operations, directly impacting both financial efficiency and patient care. Reliable forecasting of drug demand is essential for optimizing inventory management and minimizing costs. This study presents a preliminary step toward developing a comprehensive solution by leveraging Long Short-Term Memory (LSTM) networks for time series forecasting of drug demand across three types of datasets: non-stationary, intermittent, and multi-seasonal. In addition to the baseline LSTM predictions, five post-prediction strategies are introduced to enhance the accuracy of forecasts, including directional (DiL) and undirectional (UDiL) double prediction methods, as well as self-adaptation techniques such as Simple Residual Adjustment (SRA), Recursive Residual Model (RRM), and Time Adaptive Demand Forecasting Adjustment (TADFA). The results indicate significant improvements in prediction quality, particularly when employing DiL and TADFA. Furthermore, the predicted values were used to simulate the inventory management costs under a model where both overprediction and underprediction are penalized equally. This work highlights the importance of accurate forecasting and demonstrates the potential for AI-driven solutions to improve pharmaceutical inventory management in hospitals.

Keywords. LSTM (Long Short-Term Memory), Drug Demand Forecasting, Hospital Inventory Management, Time Series Prediction, Error-Correction Mechanisms

1. Introduction

First Effective pharmaceutical inventory management [1] is crucial for hospitals to ensure the availability of essential drugs while minimizing costs and reducing waste, [2]. Poor inventory management can lead to both drug shortages and overstocking, resulting in financial strain and compromised patient care [3]. Accurate drug demand forecasting is a vital component in addressing this challenge, enabling hospitals to make informed

¹ Corresponding Author: Sayan Kaennakham, sayan_kk@g.sut.ac.th.

decisions regarding drug procurement and inventory levels. Artificial intelligence (AI), particularly machine learning techniques, has shown promise in improving forecasting accuracy, offering a potential solution to this pressing issue, [4].

Long Short-Term Memory (LSTM) [5] networks have emerged as a powerful tool for time series forecasting due to their ability to capture complex patterns and dependencies in sequential data. In this study, LSTM is applied to predict drug demand across three distinct types of time series datasets [6], each reflecting different demand patterns that hospitals typically encounter. While LSTM provides strong baseline predictions, further refinement is necessary to enhance the model's accuracy, particularly in scenarios involving non-stationary, intermittent, or multi-seasonal demand patterns [7].

To address this, several post-prediction adjustment strategies are introduced to improve the quality of LSTM predictions. These include double prediction methods—directional and undirectional—along with self-adaptive strategies such as Simple Residual Adjustment, Recursive Residual Model, and Time Adaptive Demand Forecasting Adjustment. By applying these strategies, the goal is to better align drug demand predictions with actual requirements, thus optimizing inventory management. Ultimately, this preliminary work aims to pave the way for a fully automated, AI-driven system that minimizes inventory costs while ensuring the timely availability of medications in hospitals.

2. Mathematical Background

2.1 The Mathematical Model for Pharmaceutical Inventory Expenses

This preliminary work is a key step toward our long-term goal of optimizing pharmaceutical inventory expenses. Here, we present the first version of our mathematical model and highlight the critical role of accurate drug demand forecasting. Objective function: minimize cost

$$Z = \sum_{i=1}^N \sum_{t=1}^T [C_h \cdot (I_t + R_t) + C_o \cdot R_t + Cst. + Cov.] \quad (1)$$

subject to: $I_{t+1} = I_t + R_t - D_t$, $0 \leq I_t \leq I_{max}$, $I_{min} \leq I_t \leq I_{max}$, $R_t = D_t - I_t + I_{min}$.

The refilling cost model in this work contains the following parts:

Storage Cost: $C_h \cdot (I_t + R_t)$,

Medicine Cost: $C_o \cdot R_t$,

Shortage Penalty Cost: $Cst. = C_s \cdot \max(0, D_t - I_t - R_t + I_{min})$,

Overstock Penalty Cost: $Cov. = C_{ov} \cdot \max(0, I_t + R_t - I_{max})$,

with indices: i : Index for items or products, where $i = 1, 2, \dots, N$, and t : Index for time periods, where $t = 1, 2, \dots, T$.

Notations:

- Z : Total cost over all items and time periods.
- C_h : Holding (or storage) cost per unit.
- C_o : Ordering cost per unit.
- C_s : Shortage penalty cost coefficient.
- C_{ov} : Overstock penalty cost coefficient.
- I_t : Inventory level at time t .

- R_t : Replenishment quantity at time t .
- D_t : Demand at time t .
- I_{max} : Maximum allowed inventory level.
- I_{min} : Minimum required inventory level.

2.2 Long Short-Term Memory Neural Network

Long Short-Term Memory (LSTM) is a recurrent neural network (RNN) designed to model sequential data while avoiding the vanishing gradient problem. Each LSTM unit has a cell state C_t and uses three gates: forget gate f_t , input gate i_t , and output gate o_t , to control information flow. The forget gate determines how much of the previous cell state C_{t-1} to keep, defined as follows

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f). \quad (2)$$

The input gate determines what new information to store in the cell state, defined as follows

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i). \quad (3)$$

The new candidate values $\tilde{C}_t$ for the cell state are computed using the hyperbolic tangent function

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c). \quad (4)$$

Finally, the cell state is updated

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t. \quad (5)$$

The output gate o_t determines the next hidden state h_t , which influences the output

$$h_t = o_t * \tanh(C_t). \quad (6)$$

These mechanisms allow LSTM to control the flow of information effectively over long time periods, making it suitable for time series tasks.

2.3 Post-prediction Strategies

Following the application of LSTM to the three time series datasets, post-prediction strategies were explored to improve performance. Two main approaches were tested: (1) double prediction on LSTM residuals using directional (DiL) and undirectional (UDiL) methods, and (2) self-adaptation strategies, including Simple Residual Adjustment (SRA), Recursive Residual Model (RRM), and Time Adaptive Demand Forecasting Adjustment (TADFA). Brief details of each self-adaptation method are provided below.

In a self-adaptive approach for improving time series predictions, there are several strategies, each with its own mathematical formulation. Below are various strategies that could be applied to the idea of leveraging residuals to adjust and improve predictions. At time stamp t , the actual demand is denoted by D_t and the forecast demand is denoted by $\hat{D}_t$. The residual is the difference between the actual demand and the forecast demand, denoted by $\hat{r}_t$. In this work, three self-adaptive strategies are used to adjust the forecast demand, enhancing the accuracy of prediction.

1. Simple Residual Adjustment (SRA)

In this method, the prediction for the next step is corrected by adding a fraction of the residual from the previous time step. The self-adjusted prediction for the next step is as follows

$$\hat{D}_t^{new} = \hat{D}_t^{LSTM} + \alpha \hat{r}_{t-1}, \quad (7)$$

where α is a hyperparameter that controls the influence of the previous residual on the next prediction. Typically, $0 < \alpha < 1$.

2. Recursive Residual Model (RRM)

At time stamp t , the residual $\hat{r}_t$ is estimated adaptively using the corrected predictions from the two preceding time stamps. This residual $\hat{r}_t$ is then added to the LSTM prediction at time stamp t to form the new prediction $\hat{x}_t^{new}$.

Residual Adjustment: The residual for time stamp t can be estimated based on the previously corrected predictions as follows

$$\hat{r}_t = f(\hat{r}_{t-1}, \hat{r}_{t-2}), \quad (8)$$

where $f(\hat{r}_{t-1}, \hat{r}_{t-2})$ is a self-adaptive function that captures the relationship between previous residuals. In this form, the function depends recursively on the residuals, where the correction for the current time stamp is a function of the previous residual and a recursive term as follows

$$f(\hat{r}_{t-1}, \hat{r}_{t-2}) = \beta \hat{r}_{t-1} + (1 - \beta) f(\hat{r}_{t-2}, \hat{r}_{t-3}), \quad (9)$$

where β is a weight between 0 and 1. This model recursively involves older residuals, making it more sensitive to long-term dependencies in the error pattern. **Prediction Update:** The corrected prediction at time stamp t is then given by

$$\hat{D}_t^{new} = \hat{D}_t^{LSTM} + \hat{r}_t. \quad (10)$$

3. Time-Adaptive Demand Forecasting Adjustment (TADFA)

At time stamp t , the TADFA formula updates the adjustment parameter α_t based on the forecast residual of previous time stamp as follows

$$\alpha_t = \alpha_{t-1} + \eta \hat{r}_{t-1}, \quad (11)$$

where η is the learning rate (a constant). The forecast for the current period t is then adjusted as follows

$$\hat{D}_t^{new} = \hat{D}_t^{LSTM} + \alpha_t. \quad (12)$$

3. Experimental Setup

In this study, an LSTM model was used to predict values on three time series datasets, where each dataset was tested with both 80/20 and 70/30 train/test ratios. The dataset, consisting of 'Date' and 'Value' columns, was first normalized using a MinMaxScaler to scale the data between 0 and 1, ensuring that the LSTM could effectively process the values. A sliding window technique with a look-back period of 1 was used to create input-output pairs from the dataset, where each input represented the previous time step's value. The LSTM architecture comprised one LSTM layer with 4 units, followed by a Dense output layer. The model was trained for 100 epochs with a batch size of 1, using mean squared error as the loss function and the Adam optimizer to adjust the weights. Following training, predictions were generated for both the training and test datasets, and the results were inverse-transformed to their original scale for accurate comparison. The performance of the model was evaluated using the Root Mean Square Error (RMSE).

4. Main Results and Discussion

The results in **Table 1** demonstrate the performance of LSTM and five post-prediction strategies—UDiL, DiL, SRA, RRM, and TADFA—across three distinct time series datasets. For the 70/30 train-test ratio, Dataset-1 (Non-stationary time series) shows an initial RMSE of 1.4460 using LSTM, which is notably reduced by TADFA to 1.0846, marking a significant improvement. Dataset-2 (Intermittent time series) starts with an RMSE of 12.9950 for LSTM, but no substantial improvement is achieved, with UDiL and DiL increasing the error further. However, RRM reduces the RMSE slightly to 12.9663, and TADFA performs similarly at 12.9713. For Dataset-3 (Multi-seasonal time series), the RMSE starts at 5.0940 for LSTM, with all five strategies producing similar results, with RRM yielding the best improvement at 5.0884.

With the 80/20 train-test ratio, Dataset-1 shows consistent improvement, where the initial LSTM RMSE of 1.1950 is reduced by DiL to 1.0740. For Dataset-2, the initial RMSE of 12.3610 sees no improvement, with UDiL significantly increasing the error, though RRM lowers the RMSE to 12.2098, and TADFA follows closely at 12.1785. Dataset-3 sees minor enhancements, with LSTM starting at 5.1800 and TADFA reducing it slightly to 4.9654, indicating stable performance across the different strategies. The notable improvements occur mainly in Dataset-1 using TADFA, while Dataset-2 remains challenging across all methods.

Table 1. RMSE Comparison of LSTM and Post-Prediction Strategies Across Different Time Series Datasets Using 70/30 and 80/20 Train-Test Ratios.

Model	LSTM	Double Prediction		Self-Adaptation		
		UDiL	DiL	SRA	RRM	TADFA
Train/Test Ratio : 70/30						
Dataset-1	1.4460	1.2150	1.3410	1.3926	1.1742	1.0846
Dataset-2	12.9950	14.8950	13.1460	12.9858	12.9663	12.9713
Dataset-3	5.0940	6.1830	5.0950	5.0935	5.0884	5.0890
Train/Test Ratio : 80/20						
Dataset-1	1.1950	1.1410	1.0740	1.1848	1.0972	1.0745
Dataset-2	12.3610	16.3450	12.2470	12.3246	12.2098	12.1785
Dataset-3	5.1800	5.3920	4.9690	5.1803	4.9879	4.9654

Figure 1 presents a comparison of the estimated inventory expenses over 52 and 60 weeks for the LSTM model and three post-prediction strategies: DiL, TADFA, and RRM. The baseline LSTM model leads to the highest costs, with expenses reaching approximately THB 40,000 over the 52-week period and climbing to nearly THB 50,000 over 60 weeks. These high costs are a result of the equal THB 100 per unit penalty for overprediction and underprediction, highlighting the limitations of relying solely on LSTM predictions.

In contrast, the application of post-prediction strategies substantially reduces these expenses. Both DiL and TADFA show significant improvements, bringing the expenses down to around THB 30,000 over 52 weeks, representing notable savings compared to LSTM. Over 60 weeks, these strategies continue to demonstrate lower costs, maintaining more efficient inventory management, with TADFA and DiL yielding the best results. RRM, while also reducing costs compared to LSTM, shows slightly higher expenses than DiL and TADFA but remains more effective than the unadjusted LSTM. Overall, the post-prediction strategies, especially DiL and TADFA, clearly highlight the benefits of refining the LSTM predictions to achieve more cost-effective pharmaceutical inventory management in hospitals.

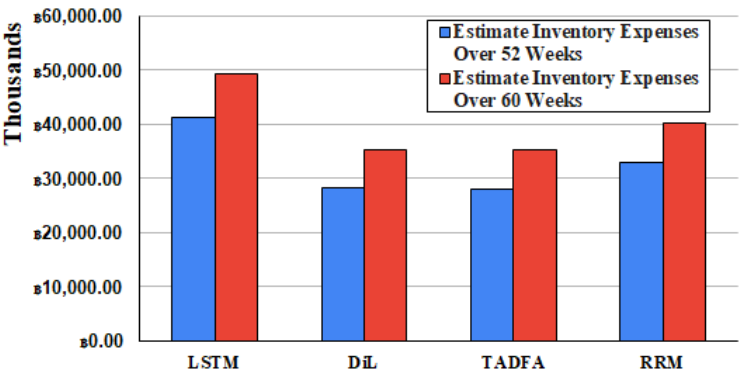


Figure 1. Comparison of Estimated Inventory Expenses Over 52 and 60 Weeks Using LSTM and Post-Prediction Strategies (DiL, TADFA, RRM)

5. Conclusions

This research has demonstrated the effectiveness of LSTM in predicting drug demand across diverse time series datasets, providing a crucial step toward optimizing pharmaceutical inventory management. The integration of five post-prediction strategies, particularly DiL and TADFA, significantly improved the quality of predictions, leading to more accurate forecasting of demand. These improvements were reflected in the reduced costs observed in the simulations, where the total expenses over 52 and 60 weeks were notably lower for models incorporating post-prediction strategies than those using LSTM alone. By showing that accurate demand forecasting can reduce both overstocking and shortages, this study underscores the critical role that AI can play in improving hospital inventory management. Future work will focus on further refining these models and developing a fully automated, AI-driven inventory management system that ensures cost efficiency and availability of essential drugs.

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